

Medicine Through Time

Master Quiz Pack

Medieval (c1250–c1500): End of Topic Quiz Pack

Instructions: Answer the 17 quick-fire recall questions below. The scrambled answers are provided in the Answer Bank on the next page.

1. What does the term 'public health' refer to?

2. Name one ancient civilization that influenced later medical ideas.

3. Why was the Catholic Church so powerful in the Middle Ages?

4. Explain how religion might both help and hinder medical progress.

5. To what extent were ancient ideas like those of Hippocrates still influential in c1250?

6. Which Ancient Greek physician originally created the Theory of the Four Humours?

7. How did the Church explain the cause of disease?

8. What was 'miasma'?

9. Why did the Church promote the ideas of Galen?

10. Explain why astrology was used to diagnose illness in the medieval period.

11. Recall: Who was Roger Bacon and what was their key contribution to medicine?

12. Name one religious treatment used in the medieval period.

13. What was the purpose of 'bleeding' or 'purging'?

14. Who were the 'flagellants'?

15. Explain the role of the apothecary in medieval medicine.

16. How effective were medieval hospitals at treating disease?

17. Recall: Who was Roger Bacon and what was their key contribution to medicine?

Medieval (c1250–c1500) Answer Bank

Ancient ideas were incredibly influential; theories like the Four Humours remained the foundation of medical education and diagnosis.

- Apothecaries mixed herbal remedies and ointments based on the knowledge passed down or from herbal manuals (Materia Medica).
- Astrology was used because people believed the alignment of planets and stars could negatively impact health, providing a supernatural explanation where science lacked answers.
- Hippocrates.
- Miasma was the belief that 'bad air' filled with foul-smelling fumes from rotting matter caused disease.
- People who whipped themselves in public to show God they were sorry for their sins, hoping to avoid disease.
- Praying, fasting, or going on a pilgrimage.
- Public health refers to the health of the population as a whole, especially as monitored and regulated by the state or government.
- Religion could help by encouraging the care of the sick (e.g. hospitals), but hinder progress by relying on supernatural explanations rather than scientific investigation.
- Roger Bacon was a 13th-Century Scientist. An English philosopher and Franciscan friar who emphasized the study of nature through empirical observation. He was imprisoned by Church leaders for suggesting that doctors should conduct their own independent experiments.
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- The Ancient Greeks or Ancient Romans.
- The Church promoted Galen because his idea that the body had a perfect design fit with the Christian belief in a single Creator.
- The Church taught that illness was a punishment from God for sin, or a test of faith.
- The Church was the central authority in Medieval Europe, controlling education, the copying of books, and holding immense spiritual and political power.
- They were ineffective at curing disease because they focused on 'care, not cure', often offering prayer and a clean resting place rather than medical intervention.
- To rebalance the Four Humours by removing excess fluids from the body.

Renaissance (c1500–c1700): End of Topic Quiz Pack

Instructions: Answer the 17 quick-fire recall questions below. The scrambled answers are provided in the Answer Bank on the next page.

1. In what year did the Black Death arrive in England?

2. What did medieval people believe caused the Black Death?

3. State one method used to try to prevent the Black Death.

4. Why did the Black Death cause such a high death toll?

5. To what extent did the Black Death change medical ideas?

6. Recall: Who was Claudius Galen and what was their key contribution to medicine?

7. What was the Renaissance?

8. What invention helped spread new medical ideas during the Renaissance?

9. Who was Thomas Sydenham?

10. Why did the influence of the Church begin to decline in the Renaissance?

11. Explain why the Theory of the Four Humours was still used despite new discoveries.

12. Who wrote 'On the Fabric of the Human Body' (1543)?

13. What did Vesalius prove about Galen's anatomical ideas?

14. What was 'transference'?

15. Explain how hospitals changed during the Renaissance.

16. Why did Vesalius' discoveries have limited immediate impact on everyday medical treatment?

17. Recall: Who was Andreas Vesalius and what was their key contribution to medicine?

Renaissance (c1500-c1700) Answer Bank

1348. · A period of 'rebirth' in art, science, and learning, beginning in the 15th century, which encouraged challenging old ideas. · A punishment from God, an unusual alignment of planets, or miasma (bad air). · An English physician known as the 'English Hippocrates', who believed in closely observing symptoms rather than relying on ancient books. · Andreas Vesalius was a Pioneering Anatomist. A Renaissance physician who published "On the Fabric of the Human Body" (1543). By performing his own human dissections, he proved over 300 errors in Galen's work (e.g., proving the human jaw bone is one piece, not two). · Andreas Vesalius. · Carrying sweet-smelling herbs, lighting fires, or running away. · Claudius Galen was a Ancient Roman Physician. Developed the Theory of Opposites based on the Four Humours. His anatomical writings, though often based on animal dissection and highly flawed, were promoted by the Catholic Church and formed the absolute basis of medical training for over 1,000 years. · Following the dissolution of the monasteries, many hospital-like institutions closed, leading to a shift towards secular funding, though they still largely focused on care rather than cure. · He proved that Galen had made over 300 mistakes because Galen had dissected animals, not humans (e.g., the human jawbone is one bone, not two). · It caused very little immediate change; the sheer scale of the disaster reinforced reliance on religion and existing theories, as physicians had no better explanations. · It remained because there was no new, comprehensive theory of disease to replace it, and physicians still wanted to offer patients explanations. · Knowing exactly where bones and organs were did not help physicians cure diseases, as they still didn't understand what caused illness. · People did not understand germs or how the disease was actually transmitted (fleas on rats), so their prevention methods were entirely ineffective. · The printing press. · The Reformation and new scientific discoveries (like the Royal Society) encouraged people to look for rational explanations rather than purely religious ones.

- The Renaissance belief that an illness or disease could be transferred to an object, animal, or another person.

18th & 19th Century Medicine: End of Topic Quiz Pack

Instructions: Answer the 18 quick-fire recall questions below. The scrambled answers are provided in the Answer Bank on the next page.

1. What did William Harvey discover about the blood?

2. What did Harvey prove wrong about Galen's theories?

3. In what year was the Great Plague?

4. Explain one way the response to the Great Plague was better than the Black Death.

5. To what extent did Harvey's discovery improve medical treatment in the 1600s?

6. Recall: Who was Andreas Vesalius and what was their key contribution to medicine?

7. What was 'Spontaneous Generation'?

8. Who published the Germ Theory in 1861?

9. What did Robert Koch contribute to Germ Theory?

10. Why did many doctors reject Germ Theory at first?

11. Explain the significance of Koch's methodology (e.g., staining microbes).

12. Recall: Who was William Harvey and what was their key contribution to medicine?

13. Who was Florence Nightingale?

14. What did Nightingale believe caused disease?

15. Name the first effective anaesthetic used in surgery (by James Simpson).

16. Explain the difference between anaesthetics and antiseptics.

17. Why did the 'Black Period of Surgery' occur after the discovery of anaesthetics?

18. Recall: Who was Roger Bacon and what was their key contribution to medicine?

18th & 19th Century Medicine Answer Bank

1665. • A nurse who dramatically improved hospital hygiene and nursing standards during the Crimean War. • Anaesthetics stop the patient from feeling pain, whereas antiseptics (like carbolic acid) kill germs to prevent infection. • Andreas Vesalius was a Pioneering Anatomist. A Renaissance physician who published "On the Fabric of the Human Body" (1543). By performing his own human dissections, he proved over 300 errors in Galen's work (e.g., proving the human jaw bone is one piece, not two). • Chloroform (1847). • He discovered that blood circulates around the body, pumped by the heart in a one-way system. • He proved that blood is not constantly manufactured in the liver and consumed by the body. • Koch successfully identified the specific microbes that caused individual diseases, such as anthrax and tuberculosis. • Koch's methods allowed other scientists to easily identify and study specific bacteria, transforming bacteriology into a rigorous science. • Louis Pasteur. • Mayors and councils took more organized action, such as ordering victims to be quarantined in their houses with a red cross painted on the door. • Microbes were everywhere, and doctors couldn't believe such tiny organisms could kill large humans; they also clung to the idea of Miasma. • Roger Bacon was a 13th-Century Scientist. An English philosopher and Franciscan friar who emphasized the study of nature through empirical observation. He was imprisoned by Church leaders for suggesting that doctors should conduct their own independent experiments. • She believed in Miasma (bad air), which drove her focus on cleaning and ventilating hospital wards. • Surgeons attempted deeper, more complex operations because patients were unconscious, but without antiseptics, infection rates and deaths skyrocketed. • The incorrect theory that rotting matter naturally created microbes or maggots. • Very little; blood transfusions were not yet possible, and his discovery did not help explain the cause of disease. • William Harvey was a Royal Physician. An English physician who discovered the full circulation of blood in the human body. By dissecting cold-blooded animals and experimenting with ligatures, he proved the heart acts as a pump, disproving Galen's theory that blood was constantly manufactured in the liver.

Modern (c1900–present): End of Topic Quiz Pack

Instructions: Answer the 23 quick-fire recall questions below. The scrambled answers are provided in the Answer Bank on the next page.

1. What disease did Edward Jenner find a vaccine for in 1796?

2. What did Jenner observe about milkmaids?

3. Who discovered the cause of cholera in 1854?

4. Explain how John Snow proved that cholera was a waterborne disease.

5. To what extent did the government support public health improvements in the mid-19th century?

6. What structure did Watson and Crick discover in 1953?

7. What is the function of DNA in relation to disease?

8. Name one modern lifestyle factor that causes disease.

9. Explain how the discovery of DNA changed the understanding of disease.

10. Why did the mapping of the human genome (completed in 2003) not lead to immediate cures?

11. Recall: Who was Roger Bacon and what was their key contribution to medicine?

12. What is a 'magic bullet'?

13. Name the first magic bullet discovered by Paul Ehrlich in 1909.

14. What was established in Britain in 1948 to improve public health?

15. Explain the impact of the NHS on medical treatment in Britain.

16. How did technology improve diagnosis in the 20th century?

17. Recall: Who was Ernst Chain and what was their key contribution to medicine?

18. Who discovered Penicillin by accident in 1928?

19. Which two scientists developed Penicillin into a usable drug?

20. What role did the US government play in the mass production of Penicillin?

21. Explain why Penicillin is considered a turning point in medicine.

22. To what extent was Fleming solely responsible for the success of Penicillin?

23. Recall: Who was Roger Bacon and what was their key contribution to medicine?

Modern (c1900-present) Answer Bank

A chemical compound that targets and kills specific disease-causing microbes in the body without harming the rest of the patient. · Alexander Fleming. · DNA carries genetic information; mutations or inherited genetic defects can cause hereditary diseases. · Ernst Chain was a Biochemist. A brilliant German-Jewish refugee who worked alongside Howard Florey at Oxford. Chain successfully discovered the complex chemical method required to extract and purify Penicillin from the mould so it could be safely tested on mice and humans. · Fleming made the initial observation, but he failed to develop it. Florey, Chain, and US funding were essential to turning it into a life-saving drug. · He mapped out cholera deaths in Soho and traced them all to a single infected water pump on Broad Street, proving it wasn't caused by miasma. · He noticed that milkmaids who had caught cowpox never caught the deadly smallpox. · Howard Florey and Ernst Chain. · Identifying the genes responsible for a disease is only the first step; creating gene therapies to fix or replace them is vastly more complex. · Initially, they maintained a 'laissez-faire' (hands-off) attitude, but the Great Stink (1858) and Snow's findings eventually forced them to act, such as building the London sewer system. · It allowed scientists to finally understand conditions that were neither infectious nor caused by lifestyle, such as Down's Syndrome or Cystic Fibrosis. · It made healthcare free at the point of delivery, dramatically improving access to treatments and hospitals for the working classes. · It was the first true antibiotic, capable of curing a wide range of bacterial infections that were previously fatal, saving millions of lives. · John Snow. · Roger Bacon was a 13th-Century Scientist. An English philosopher and Franciscan friar who emphasized the study of nature through empirical observation. He was imprisoned by Church leaders for suggesting that doctors should conduct their own independent experiments. · Roger Bacon was a 13th-Century Scientist. An English philosopher and Franciscan friar who emphasized the study of nature through empirical observation. He was imprisoned by Church leaders for suggesting that doctors should conduct their own independent experiments. · Salvarsan 606 (used to treat syphilis). · Smallpox. · Smoking, poor diet (obesity), or lack of exercise. · Technologies like X-rays, MRI scans, and blood tests allowed doctors to diagnose internal issues accurately without relying solely on patient symptoms or exploratory surgery. · The double helix structure of DNA. · The National Health Service (NHS). · They funded the mass production of Penicillin so it could be used to treat Allied soldiers during World War II.

Western Front: End of Topic Quiz Pack

Instructions: Answer the 30 quick-fire recall questions below. The scrambled answers are provided in the Answer Bank on the next page.

1. What lifestyle choice was definitively linked to lung cancer in the 1950s?

2. Name one modern technology used to diagnose lung cancer.

3. How does the government try to prevent people from smoking?

4. Explain why lung cancer is so difficult to treat effectively.

5. Compare the government's response to lung cancer with its response to cholera in the 19th century.

6. Recall: Who was Edward Jenner and what was their key contribution to medicine?

7. When did the First World War begin and end?

8. What was the 'Western Front'?

9. Name the three major battles fought by the British on the Western Front.

10. Explain the impact of the terrain at Ypres on the soldiers.

11. Why was the casualty rate so high on the Western Front?

12. Recall: Who was Joseph Lister and what was their key contribution to medicine?

13. What was the purpose of the duckboards in the trenches?

14. Name the pattern in which trenches were dug.

15. What was 'No Man's Land'?

16. Explain why the zig-zag pattern of trenches was crucial for survival.

17. To what extent did the trench environment itself cause casualties?

18. Recall: Who was Francis Crick and what was their key contribution to medicine?

19. What is Trench Foot?

20. Which weapons caused the most wounds on the Western Front?

21. Why were wounds highly prone to infection on the Western Front?

22. Explain how gas attacks affected the soldiers.

23. How effective were the preventative measures against gas attacks?

24. Recall: Who was Florence Nightingale and what was their key contribution to medicine?

25. What does RAMC stand for?

26. What was the role of the FANY?

27. Describe the 'Chain of Evacuation'.

28. Explain the importance of the Casualty Clearing Stations (CCS).

29. Why was the work of the RAMC so dangerous?

30. Recall: Who was James Simpson and what was their key contribution to medicine?

Western Front Answer Bank

1914 – 1918. • A painful condition caused by standing in cold water and mud for long periods, which could lead to gangrene and amputation. • A zig-zag pattern. • By banning advertising, increasing taxes on cigarettes, and enforcing plain packaging. • CT scans, PET scans, or bronchoscopy. • Edward Jenner was a Pioneer of Vaccination. An English country doctor who discovered the first successful vaccine in 1796. After observing that milkmaids who caught cowpox never caught smallpox, he deliberately infected a boy with cowpox to successfully immunize him against the deadly smallpox virus. • Florence Nightingale was a Founder of Modern Nursing. A legendary nurse who drastically reduced death rates at the Scutari hospital during the Crimean War (1854) by implementing strict hygiene and sanitation standards. She later published "Notes on Nursing" and established professional training for nurses. • Francis Crick was a Molecular Biologist. Working together at Cambridge University, they successfully published the double-helix structure of DNA in 1953. Their monumental breakthrough finally allowed scientists to understand how genetic traits and hereditary diseases are passed down from parents to children. • Gas weapons (like Chlorine, Phosgene, and Mustard Gas) caused temporary blindness, violent coughing, suffocation, and severe internal and external blisters. • High-explosive artillery shells and shrapnel. • Initially poor (soldiers used urine-soaked rags), but the introduction of proper gas masks (like the box respirator in 1916) made gas attacks far less lethal. • It is often not diagnosed until it has advanced to a late stage because the early symptoms are easily mistaken for minor ailments. • It stopped enemy artillery blasts from travelling down the entire length of the trench and made it harder for enemy soldiers to shoot straight down the line if they invaded. • James Simpson was a Pioneer of Anesthetics. A Scottish obstetrician who discovered the anesthetic properties of chloroform in 1847. He famously experimented on himself and his friends to find a deeper, more effective pain relief than ether, revolutionizing surgery. • Joseph Lister was a Pioneer of Antiseptic Surgery. A British surgeon who applied Pasteur's Germ Theory to surgery in 1865. By spraying carbolic acid directly onto wounds and surgical instruments, he destroyed the bacteria causing infection, drastically reducing surgical mortality rates. • Royal Army Medical Corps. • Smoking tobacco. • Stretcher bearers and medical officers had to operate in 'No Man's Land' and front-line trenches under heavy enemy fire to retrieve and treat the wounded. • The combination of static trench warfare and devastating new weapons, such as machine guns and artillery, resulted in unprecedented numbers of casualties. • The dangerous, open ground between the Allied and German front-line trenches. • The environment was deadly; freezing conditions, constant dampness, and poor sanitation led to thousands of casualties from frostbite, Trench Foot, and infectious diseases. • The First Aid Nursing Yeomanry primarily drove ambulances and provided emergency first aid. • The ground was extremely muddy and prone to flooding, which severely hampered movement and contributed to conditions like Trench Foot. • The line of trenches stretching from the English Channel to the Swiss border where the majority of fighting took place. • The soil was heavily fertilized with manure, containing high levels of tetanus and gas gangrene bacteria, which were carried deep into wounds by shrapnel. • The system of moving wounded soldiers away from the frontline: Regimental Aid Post (RAP) -> Advanced Dressing Station (ADS) -> Casualty Clearing Station (CCS) -> Base Hospital. • They were the first large, well-equipped medical facilities close to the front, where emergency surgery (like amputations to stop gangrene) was performed to save lives. • To keep soldiers' feet out of the mud and water to prevent Trench Foot. • Unlike the laissez-faire attitude of the early 19th century, modern governments actively intervene through legislation and public health campaigns to prevent lung cancer. • Ypres, the Somme, and Arras.